SURIYAPRAKASH S

Data Scientist | Machine Learning Engineer | NLP & AI

+91 7868940106 | suriyaprakash112001@gmail.com | Chennai, India | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with hands-on experience in Python, SQL, Machine Learning, NLP, and Streamlit. Built end-to-end ML applications for healthcare, HR analytics, and sentiment analysis, with expertise in predictive modeling, data visualization, and transformer-based NLP. Completed Advanced Programming Professional & Master Data Science at GUVI (IIT-M affiliated, 2025–2026).

TECHNICAL SKILLS

Languages	Python, SQL
Libraries	Pandas, NumPy, Scikit-learn, TensorFlow, Keras
NLP	Hugging Face, Transformers, TF-IDF, NLTK, SpaCy
Visualization	Matplotlib, Seaborn, Plotly, Streamlit
Cloud & Tools	AWS, Git, GitHub, VS Code, Jupyter Notebook
Databases	MySQL, SQL
Competencies	Feature Engineering, EDA, Predictive Analytics, Model Deployment, Data Wrangling, Statistical Analysis, Classification, Regression, Model Evaluation

PRACTICAL EXPERIENCE

Machine Learning Projects – Self-Initiated & Academic — Jan 2024 – Present

Independent Developer | Chennai, India

- Built 6 end-to-end ML applications spanning healthcare AI, HR analytics, NLP, and agriculture data analysis.
- Deployed interactive dashboards using Streamlit for real-time predictions and data-driven insights.
- Developed NLP and transformer-based solutions achieving up to 92% classification accuracy.
- Applied feature engineering, EDA, and model evaluation across diverse datasets including clinical, HR, and agriculture data.

KEY PROJECTS

Multilingual Medical Support Chatbot GitHub ↗

Python | LLM | NLP | Transformers | Hugging Face | Streamlit

- Built a transformer-based medical chatbot using Hugging Face models to process multilingual health queries and generate context-aware responses.
- Reduced model inference latency by 30% via efficient tokenization and batching strategies.
- Deployed a Streamlit multilingual interface supporting regional languages, improving healthcare accessibility.

Employee Attrition Analysis & Prediction GitHub ↗

Python | Scikit-learn | Pandas | XGBoost | EDA | Streamlit

- Built a Random Forest and XGBoost classification pipeline achieving 87% accuracy on IBM HR dataset.
- Engineered features and performed EDA to identify major attrition drivers for business decision-making.
- Developed a Streamlit dashboard for real-time employee attrition risk scoring.

Multiple Disease Prediction System GitHub ↗

Python | Scikit-learn | Pandas | Pickle | Streamlit

- Built a clinical AI system predicting Kidney Disease (CKD), Liver Disease, and Parkinson's Disease using separate trained ML pipelines with confidence scoring.
- Engineered input alignment logic handling feature mismatches across 3 disease models for reliable real-time inference.
- Deployed a multi-page Streamlit dashboard with disease-specific input forms and probability-based result display.

AI Echo – ChatGPT Review Sentiment Analysis GitHub ↗

Python | NLP | Scikit-learn | TF-IDF | Streamlit

- Developed an NLP sentiment classifier (Positive / Neutral / Negative) using TF-IDF vectorization achieving 92% accuracy.
- Built an interactive Streamlit dashboard with real-time text preprocessing, sentiment prediction, and trend visualization.

SecureCheck – Police Check Post Digital Ledger GitHub ↗

Python | SQL | Streamlit | Pandas

- Architected a checkpoint analytics system integrating SQLite for vehicle stop record management and SQL-driven insight extraction.
- Built dashboards to analyze traffic patterns and violations, demonstrating data engineering and BI skills.

AgriData Explorer – Indian Agriculture Data Analysis GitHub ↗

Python | Pandas | Matplotlib | Seaborn | Plotly

- Conducted EDA on ICRISAT district-level data; built visual reports comparing crop production trends across Indian states and years.
- Delivered Plotly interactive dashboards with dynamic filtering and drill-down analysis for domain stakeholders.

ACHIEVEMENTS

- Built 6+ production-ready ML applications spanning healthcare AI, NLP, HR analytics, and agriculture.
- Achieved 92% accuracy in NLP-based sentiment classification using TF-IDF and ensemble models.
- Reduced medical chatbot inference latency by 30% through optimized tokenization and batching.
- Developed a multi-disease clinical AI system covering 3 disease domains with real-time confidence scoring.

EDUCATION

Advanced Programming Professional & Master Data Science 2025 – 2026

GUVI (IIT-M Affiliated), Chennai

B.Tech in Information Technology 2021 – 2025

TJ Institute of Technology, Chennai

CGPA: 7.85 / 10

Relevant Coursework: Machine Learning, Deep Learning, DBMS, Data Mining, Data Structures, Statistics

Higher Secondary Certificate (12th) 2019

Kamalammal Matric Higher Secondary School

449 / 600 (74.8%)

Secondary School Certificate (10th) 2017

E.R.K. Higher Secondary School

460 / 500 (92%)

RELEVANT COURSEWORK

Machine Learning • Deep Learning • Natural Language Processing • Statistics & Probability • Data Structures & Algorithms • Database Management Systems • Data Mining • Computer Vision

CERTIFICATIONS

- Advanced Programming Professional & Master Data Science – GUVI (2025–2026)
- Python for Data Science & Machine Learning – GUVI
- SQL for Data Analytics – GUVI

LANGUAGES

English (Professional Working Proficiency) **Tamil** (Native)